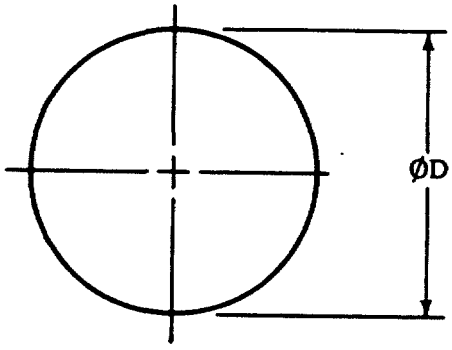
MILITARY SPECIFICATION SHEET
BALLS, BEARING, BRONZE, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the bearing described herein shall consist of this specification and the latest issue of DOD-B-82737.



NOTES:

1. All dimensions are in millimeters. Column headings in tolerance tables are in accordance with AFBMA-STD-10.
2. For tolerances by grade see Table II for individual balls and Table III for lots of balls. Tolerances are in micrometers (μm).
3. Grade 10 balls are preferred in most military applications.
4. Referenced government (or nongovernment) documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

FIGURE 1. BALLS, BEARING, BRONZE, METRIC.

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TABLE I. Dash numbers and dimensions.

ØD NOM	GRADE 200	GRADE 500	GRADE 1000
	DASH NUMBERS		
1.5	20015	50015	100015
2	20020	50020	100020
2.5	20025	50025	100025
3	20030	50030	100030
3.5	20035	50035	100035
4	20040	50040	100040
4.5	20045	50045	100045
5	20050	50050	100050
5.5	20055	50055	100055
6	20060	50060	100060
6.5	20065	50065	100065
7	20070	50070	100070
7.5	20075	50075	100075
8	20080	50080	100080
9	20090	50090	100090

ØD NOM	GRADE 200	GRADE 500	GRADE 1000
	DASH NUMBERS		
10	200100	500100	1000100
11	200110	500110	1000110
11.5	200115	500115	1000115
12	200120	500120	1000120
13	200130	500130	1000130
14	200140	500140	1000140
15	200150	500150	1000150
16	200160	500160	1000160
17	200170	500170	1000170
18	200180	500180	1000180
19	200190	500190	1000190

REQUIREMENTS:

1. Material: Bronze, non-ferrous conforming to the chemical composition of UNS C46400 (CDA464) in accordance with QQ-B-637 and as specified in procurement document.

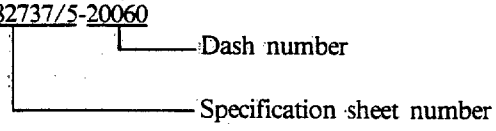
2. Hardness: Shall be 75-98 HRB or equivalent.

3. Surface Roughness: Tolerance limits as specified in Table II and in accordance with ANSI B46.1.

4. Material Density: Shall be 8.415 grams per cubic centimeter.

5. Part Number: Consists of the letter "D", the basic number of this specification sheet and a dash number taken from Table I.

Example: D82737/5-20060



D82737/5-20060 Indicates a grade 200 bearing ball of bronze with a basic diameter of 6 millimeters.

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TABLE II. Tolerances by grade for individual balls.

GRADE	ALLOWABLE BALL DIAMETER VARIATION 1/	ALLOWABLE DEVIATION FROM SPHERICAL FORM 1/	MAXIMUM SURFACE ROUGHNESS ARITHMETICAL AVERAGE 1/
	V_D	W	
200	5	5	0.2
500	13	13	— 2/
1000	25	25	— 2/

1/ Tolerances in micrometers (μ m)
2/ Not applicable

TABLE III. Tolerances by grade for lots of balls.

GRADE	ALLOWABLE LOT DIAMETER VARIATION 1/	BASIC DIAMETER TOLERANCE 1/
200	10	±25
500	25	±50
1000	50	±125

1/ Tolerances in micrometers (μ m)

Custodians:
Army - AT
Air Force - 99

Review activities:
Navy - SH
Air Force - 84
DLA - IS

Preparing activity:
Navy - OS

Agent:
DLA - IS

(Project 3110-0737-05)

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